Canonical BP: v1 and v2

Flooding sum-product for binary BP and four-state BP4.

$$\mathrm{clip}_L(x) = \max(-L, \min(L, x)), \qquad \varepsilon = 10^{-10}, \qquad d = 0.1$$

BP4 check inputs are commute/anticommute projections; $\chi_{cv}(a) = 1$ when a anticommutes with the local check Pauli.

v1

SUM-PRODUCT CHECK

$$a \boxplus b = 2\mathrm{atanh}[\tanh(a/2)\tanh(b/2)]$$

$$\hat{m}_{c\to v}^{(t+1)} = (-1)^{s_c} \boxplus_{25}^{v' \in \partial c \setminus v} m_{v'\to c}^{(t)}$$

BINARY VARIABLE

$$m_v^{(t+1)} = \mathrm{clip}_{25}(\lambda_v + \sum_{c \in \partial v} \hat{m}_{c\to v}^{(t+1)})$$

$$m_{v\to c}^{(t+1)} = \mathrm{clip}_{25}(m_v^{(t+1)} - \hat{m}_{c\to v}^{(t+1)})$$

BP4 VARIABLE

$$m_v^{a,(t+1)} = \mathrm{clip}_{25}(\lambda_v^a + \sum_{c \in \partial v} \chi_{cv}(a)\hat{m}_{c\to v}^{(t+1)})$$

$$m_{v\to c}^{a,(t+1)} = \mathrm{clip}_{25}(m_v^{a,(t+1)} - \chi_{cv}(a)\hat{m}_{c\to v}^{(t+1)})$$

The subscript 25 on box-plus means an ordered fold with inputs and each pairwise result clipped to magnitude 25. BP4 may stop at iteration zero; nonidentity ties favor X, then Y, then Z.

v2

SUM-PRODUCT CHECK

$$q_{cv}^{(t)} = (-1)^{s_c} \prod_{v' \in \partial c \setminus v} \tanh(m_{v'\to c}^{(t)}/2)$$

$$\hat{m}_{c\to v}^{(t+1)} = 2\mathrm{atanh}[\mathrm{clip}_{1-\varepsilon}(q_{cv}^{(t)})]$$

BINARY VARIABLE

$$m_v^{(t+1)} = \lambda_v + \sum_{c \in \partial v} \hat{m}_{c\to v}^{(t+1)}$$

$$m_{v\to c}^{(t+1)} = d\, m_{v\to c}^{(t)} + (1-d)(m_v^{(t+1)} - \hat{m}_{c\to v}^{(t+1)})$$

BP4 VARIABLE

$$m_v^{a,(t+1)} = \lambda_v^a + \sum_{c \in \partial v} \chi_{cv}(a)\hat{m}_{c\to v}^{(t+1)}$$

$$m_{v\to c}^{a,(t+1)} = d\, m_{v\to c}^{a,(t)} + (1-d)(m_v^{a,(t+1)} - \chi_{cv}(a)\hat{m}_{c\to v}^{(t+1)})$$

Only the check product is clamped at magnitude 1 - epsilon. Finite priors and messages have no global clip. BP4 completes one sweep before stopping and uses the strict I, X, Z, else Y decision rule.

Binary decision: bit 1 iff posterior LLR < 0; exact zero selects 0.